

Case study 1-Tyre Road Wear Particles

Context: Designing a passenger car tyre that improve tyre wear properties and minimise pollution.

Production of tyres: total is around 17 million tons of tyres / year.

Total emission of wear debris from tyres = around 20% x 17 million tons of tyres/year (overall more waste produced by the tyres after manufacturing it).

Life-cycle of Tyres: After use, TRWP end up in the atmosphere, on the road, in rivers, road residue, sludge, incinerated or recycled.

Problem: Tyres are one of the main ways pollutions is caused; by realising tyre road wear particles.

Cause: TRWP are caused due to friction between vehicle tyres and road surface. This is proven by seeing a mixture of tyre tread material and pavement material on the road surface.

What are TRWP: cigar shaped particles, generally below 1mm in length, typically between 70-100µm, with average density of 1.8. TRWP are a mix of **TP** (Tread particle) coming from the tyre tread, and are rich in polymers, carbon black, and manufacturing additives like zinc and sulphur. And, **RP** (Roadway Particle) generated from road surface, these contain high levels of minerals and hazardous polycyclic hydrocarbons (PAHs).

TRWP Hazards: TRWP are a blend of road and tyre materials. They contain potentially harmful compounds, such as **PAHs (Polycyclic Aromatic Hydrocarbon)** and **heavy metals (zinc, when used during vulcanisation)**.

Tyre Wear particles leachate (liquid that results when water comes into contact with TRWP and extracts soluble substances from them): For humans, both TP and TRWP didn't cause cell damage or oxidative stress, but they did cause inflammation that got worse with higher concentrations—shown by increased TNF-α levels (tumour necrosis factor).

Future research will look for 6PPD-quinone and other chemicals from tyre particles in water and start testing more fish and other animals to see if they are harmed. (6PPDQ is toxic for fish).

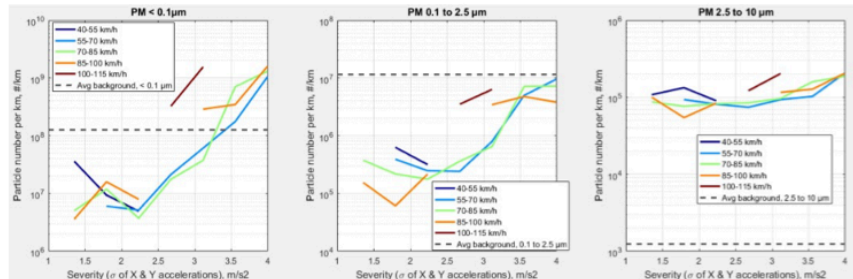
Size and volume effect of TRWP: Small particles have the largest count (important for inhalation study, as they are tiny enough to enter the lungs).

Weight% of the small particles is almost negligible (this is important for leaching as it means they don't leach many chemicals into water, so less overall impact for pollution in rivers or soil compared to bigger particles).

The **design** and **material selection** in tyres is strongly influence the toxicity and pollution protentional of TRWP.

Effect of driving behaviour on size of TRWP:

Same conclusion if we look at the different sizes : severity is much more important than speed



- Tire wear is proportional to Friction Energy :

$$\text{Wear} \cong \text{Friction Energy} = \text{Friction Force} \times \text{Sliding distance}$$

$$\text{Friction Force} = \text{Mass}_{\text{vehicle}} \times \text{Acceleration of vehicle}_{\text{longitudinal \& lateral}}$$

Depends on Tire Design
& Friction Force

Fundamental Questions:

1) Design: Is abrasive or smear wear better?

Smear wear is better as it produces lower severity friction, showing no abrasion pattern (smearing).

2) Which materials and polymer blend and which fillers or other additives are required?

For rubber compound: **FSBR**(Functionalised SBR) and **NR**(natural rubber) are the great combo for wear resistance, wet traction, energy loss, tear resistance, and heat degradation resistance.

6PPD as antioxidant (protect against thermal aging), **minarex oil** (for better wear resistance, rolling resistance and longevity) as plasticiser. **Silica** (improves rolling resistance and helps reduce fuel economy and not oil-based) and **Carbon black** (enhances wear resistance, durability and dry traction) as fillers.

3) How would you prepare the rubber compound for moulding into a car tyre?

Testing method/Protocol:

This is used to test for how easily they deform and wear, resistance to friction, and cracks.

Step 1: Analyse viscoelastic properties (how the rubber stretched and return under force)

Step 2: Prepare rubber samples (shape them to fit into a custom test rig that mimics tyre-road contact)

Step 3: Apply abrasion using UDM(uni-directional dynamic abrasion), this step simulates road wear by applying rotation and pushing forces on the rubber.

Test Rig Components (Middle section of tyre)

- Blade Mount and Removable Blade: Cuts or presses into the rubber like a road surface would.
- Bi-axial actuator: Moves the blade up/down and rotates it to stimulate tyre movement.
- Then the rubber sample is tested (by putting powder or smt and then seeing smear patterns)

Effect of crosslinks and Carbon black:

Too much crosslinking = rubber breaks too soon

Carbon black = helps rubber last longer

Best performance =CBH (carbon black, high level)

4) Explain how your tyre would perform in other measures such as fuel economy or emergency braking tests?

The use of FSBR and optimised curing helps reduce rolling resistance, meaning less energy to roll. Improving fuel economy and reduces CO2 emission.

The high toughness and dual-phase structure (NR+filled FSBR) improve grip and energy absorption during sudden stops. This gives shorter braking distances, and especially in dry and moderately wet conditions.

Link to solution ppt:

https://qmplus.qmul.ac.uk/pluginfile.php/4312042/mod_resource/content/2/%231-Keizo%20AM%20Solution.pdf

Model answers:

NOTE: Tyre tread compounds that solves TRWP issues keeping the other tyre performances

1) Desing a tyre tread compound that remove materials which are harmful to human and environment.

Use a mix of FSBR and NR, with greener alterative additives such as soya oil (bio-based plasticisers)., and greener oxidants such as hydrogen peroxide or natural antioxidants like tocopherols, this compound eliminates toxic additives such as polycyclic aromatic oils and uses sustainable, bio-based alternatives to reduce health and environmental impact while maintaining performance,

2) Design high toughness compound enhancing smearing to reduce TRWP.

High toughness is done through an optimised curing system with a higher crosslink density, enhancing energy dissipation and resistance to mechanical wear. The use of a dual-phase structure, where unfilled NR provides elasticity and highly filled FSBR contributes stiffness, results in a bi-continuous morphology that resists crack

propagation and promotes smearing under friction. Minimising generation of tyre and road wear particles, contributing to both durability and sustainability.

3) Design a tyre tread compound which has bi-continuous structure of unfilled NR and highly filled SBR to achieve high toughness.

There are two factors that contradict, them being:

- High loading (Wear resistance vs Rolling Resistance graph) giving better wear and saturation of energy loss/WET grip.
- Less loading (Wear resistance vs Rolling resistance graph) giving better energy loss and saturation of wear.

This trade-off can be achieved by a bi-continuous polymer blend structure which utilises highly filled SBR phase, by using functionalised SBR with chain end-modification. This structure optimises two extreme trade-off performances.

Things you should know model answers:

1) What type of abrasion take place in practice?

Smear and abrasion wear

2) What is the difference between smear wear and abrasive wear?

Abrasive wear: when material is removed due to hard particles or rough surfaces, scratch and cut into the tyre rubber.

Smear wear: less severe and occurs when rubber deforms and 'melts' into the road due to friction, appearing as smearing on the tread surface.

3) Which wear process is best for tyre life?

Smear wear as it causes less material loss and reduces the emission of TRWP, extending the tyre's usability and minimises environmental impact.

4) How are these behaviours measured using lab tests?

Step 1: Prepare rubber compound

Step 2: Use biaxial actuators and a cutting blade to simulate tyre-road contact

Step 3: Measure abrasion patterns and material loss

This test allows observation of deformation (smear) vs. material removal (abrasion)

5) How does the impact of this relate to rolling resistance and wet grip?

Lower rolling resistance (achieved by FSBP and optimal cross-linking/carbon black) improves fuel economy and lowers emissions.

Wet grip: is enhanced by the combination of NR and FSBP, providing better energy absorption and traction during emergency braking.

6) What should the ideal compound look like and what does each ingredient add?

Polymers: FSBR(improved wet grip, wear resistance) + NR for energy absorption and tear resistance.

Fillers: Carbon black (durability, reinforcement) , silica (wet traction).

Additives: 6PPD (antioxidant, reduces oxidative degradation), minarex oil (plasticiser for flexibility) and zinc (vulcanisation agent but watch for environmental toxicity)

7) If you use a blend of compounds how should the materials appear in their microstructure and why?

The microstructure should show homogenous dispersion of fillers, and polymers to avoid stress concentration points.

A dual-phase structure with FSBR and NR allows a balance of toughness and flexibility, enhancing wear and braking performance while reducing TRWP generation.

8) How are tyres made?

Compounding: Raw materials (polymers, fillers, additives) are mixed

Shaping: Compounds are formed into tread, sidewall, and other parts

Assembly: Parts are assembled into a 'green' uncured tyre

Curing/Vulcanisation: Tyre is heated in a mould to create crosslinks, giving it its final shape and mechanical properties.

Testing: manufacturers carry out rigours testing before the tyre hits the road.

Case Study 2–Aircraft Track Fairing

What Aircraft Fairing do:

- 1) Provides aerodynamic cover to reduce drag
- 2) Lightweight
- 3) Protect the flap track mechanism
- 4) Resist conditions at altitude of 42,000 ft.
- 5) It is an L shape structure

Constraints:



Property	In-service component value
Mass of final component	3.40kg
Main material mass	3.02kg
Deflection at design load (stiffness requirement)	1.66mm

Objective:

Produce a study of your chosen materials for the Aircraft Track Fairing

- Which manufacturing process to select? What are the steps involved?
- How many different materials needed to manufacture this component?
- What would you do with 25% wasted material?
- Investigate an end-of-life route for this component after its 25-year service life.

Manufacturing Process:

Autoclave cure - a sandwich panel structure layer

- Sandwich panel structure with ply-oriented layup:
 - 1) Each skin has three layers at 0, 90 and 45 deg with the fibre direction
 - 2) Each layer is degassed by vacuum before next layer is applied
 - 3) Tool 'on site' at GKN used for hand lay-up for (this 3 secondary structure)
 - 4) Autoclave cure with internal vacuum applied also
 - 5) Trim excess (overlay is cut off)

Global Warming Potential

As a secondary component, the Track Fairing has been highlighted as a potential component that can satisfy the push toward economically viable design strategies with the lowest environmental impact for aerospace...

Can you consider the carbon footprint (calculated in Global Warming Potential GWP kg CO² eq.) for the materials you select?

Table of useful material GWP values		
Name	Reference flow & unit	CML2001 - Aug. 2016, Global Warming Potential (GWP 100 years) [kg CO ₂ eq.]
Aluminium profile (EN15804 A1-A3)	1kg	8.678598316
Concrete C30/37 (Ready-mix concrete) (EN15804 A1-A3)	1kg	0.111628227
Flax	1kg	-1.3860008
Carbon fibre (UD) PAN-based	1kg	30.5861139

End-of-life cycle problem:

- The composite manufacturing process produces un-cured offcut waste from high value material, typically 10-35% of material purchased.
- Most composite manufacturing waste ends in **landfill** or **incineration**
- Value retention of waste is minimal
- **Current waste recovery has low value retention**
- Decarbonising use-phase of products through **alternative fuel sources** will **not hit Net Zero** targets alone
- Big changes to existing systems can be complicated and costly
- Many sustainable technologies are not cost-effective, even at scale
- Current disposal route: **incineration and/ or landfill**

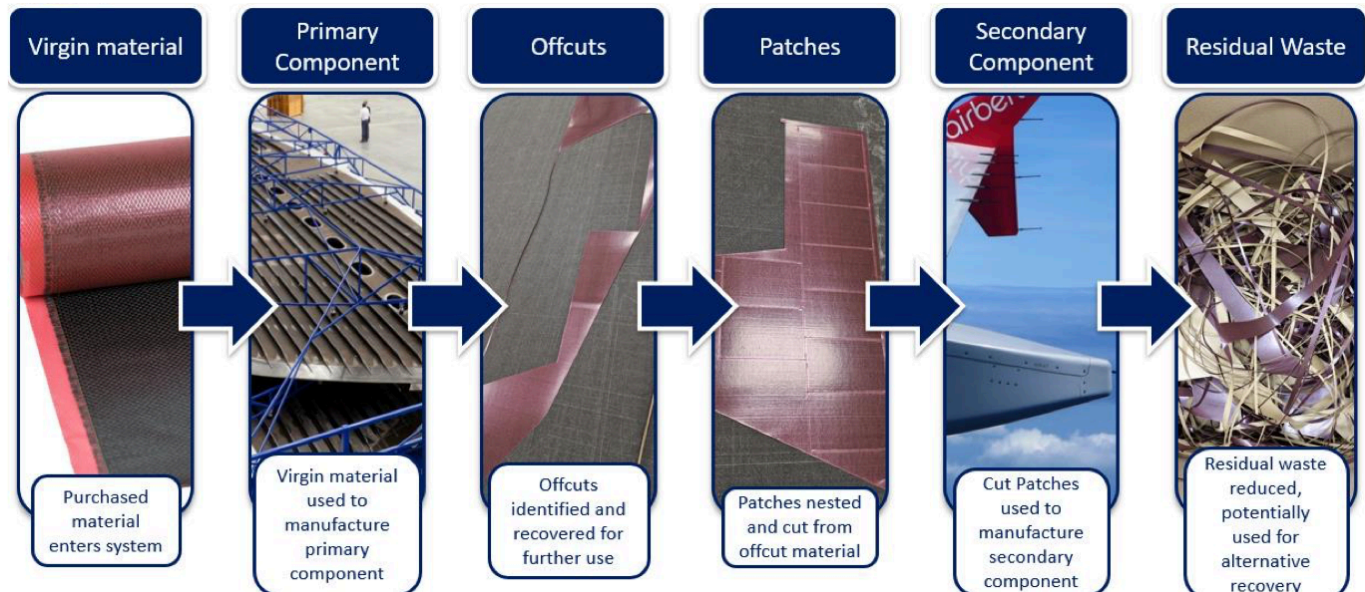
Ways to decarbonise (without changing type of fuel):

- Decarbonising CF (carbon fibre) production
- Production and EOL (end-of-life) waste

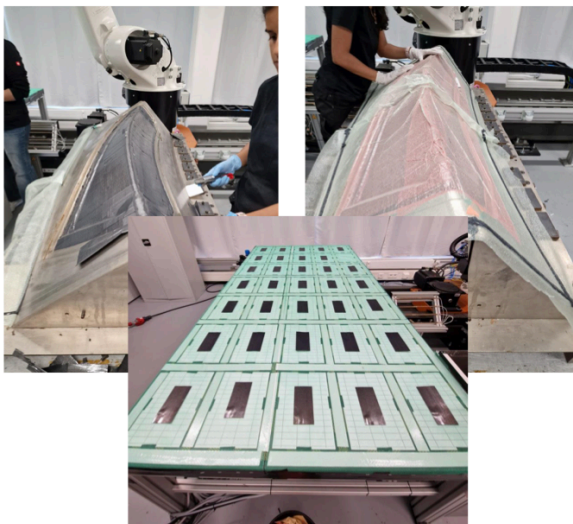
What is required to begin looking at circular solution?

- Maximise value retention of waste fibre fabrics by retaining fibre length
- Ensure overall improvement in sustainability, considering all manufacturing steps
- Maintain or reduce any modified component's mass
- Minimise or eliminate impact on production rates of existing components
- Be cost effective for a viable business case

This can be solved by using patches:



Patches to Parts



- Cevotec
- Patch size and cutting
- Logistics of prepreg storage and transportation
- Can this material be qualified as virgin rather than recycled or would it have to be redirected to replace internal glass applications – qualification difficult!

Things you should know–Model answers:

1) What does the fairing do in service?

It is a flap track that allows for contraction and expansion and is a secondary component.

- Provides an aerodynamic cover over the flap track, allowing contraction and expansion, acts as a lightweight, protective secondary structure.

2) How does the L section of the fairing connect to wing?

- It attaches structurally to the wing via the L-shaped profile, providing strength and aerodynamic integration.

3) How you choose the materials for the fairing, what are we looking to optimise?

- Optimise for lightweight (making it lightweight, provides a significant improvement in energy efficiency in the overall plane), high strength, corrosion resistance, durability at altitude, and manufacturability using composite sandwich structures.

4) What are the requirements?

- Reduce drag, be lightweight, protect flap mechanism, resist conditions at 42,000 ft, maintain structural integrity with L-shape.

5) How can we increase the flexural stiffness whilst keeping the mass low?

Use sandwich panel structures with ply-oriented and autoclave curing to maximise stiffness-to-weight ratio.

6) Which materials are best for this application?

- Carbon fibre reinforced composites with sandwich panel structure (carbon fibre skins and lightweight core).

7) How you make them?

- Hand lay-up of plies with vacuum degassing, followed by autoclave curing, then trimming excess material.

8) How might you test them?

- Perform flexural stiffness tests, environmental exposure tests at altitude conditions, fatigue testing, and dimensional accuracy checks.

9) How can you exploit waste/recycle stream materials to make them?

- Reuse uncured offcuts by maximising fibre length retention, repurpose waste composites for secondary components or recycled fibres, focusing on sustainability without increasing mass or production time.

Case Study 3 Advances in Interbody Cages for Spinal Fusion

Challenge:

- Design a cage concept with an adaptive modulus to allow the new that has grown into the pores of the cage to bear an increasing load over time.
- Typically occurs in elderly people, where collagen starts to break down.
- The discs are made up of two different types of collagens, but when it starts to shrink and lose fluid. It puts a lot of pressure on the nerves.

Goal:

- To design a cage that can overcome the limitations of existing cages. The limitations being usage of materials such as PEEK, which are bioinert (don't bond well with bone), impact (delay or incomplete bone growth). Usage of Ti (Ti carries most of the bearing load, reducing the mechanical stimulus needed for bone growth, causing bone resorption and weakening).
- Cage being too stiff or small, resulting in sinkage into the vertebral endplates.
- The cage size may not perfectly fit a patient's unique anatomy.
- Radiolucency: Ti can obscure post-operation images during CT scan.

Solution:

- 3D printed Ti outer to sit on the Apophyseal Ring.
But this is expensive, issue with supply chains, but beneficial for patient as the cage will be made in regard to their bone anatomy.
- Open porous structure with concave pores and submicron surface features. This allows for bone tissue growth into the cage (porosity).
Concave pores mimic natural bone structure, encouraging attachment and activity.
Resulting in better fusion between the cage and native bone, reducing risk of non-union.
- Slot-in inner part (like a removable insert) made from magnesium alloy that can gradually dissolve in the body (resorbable).
- This insert can be customised at the time of the surgery to control how fast it dissolves depending on the patient's needs.
- Since Mg reacts quickly in the body, it's likely to be coated with chitosan (a natural, body-friendly material) to control the reaction rate and protect it.
- The cage has a space in the middle, where the surgeon can add bone graft material (material used to aid bone growth) during surgery. This can be substances like rhBMP-2, which help the bone grow and heal faster.

Key design considerations:

1) Endplate contact size: Is large or small better relative to the risk of subsidence?

- Larger endplate (bony and cartilaginous layer) contact size is better, as the contact area distributes mechanical loads more evenly, reducing stress on the vertebral endplates and thereby minimising the risk of subsidence

(sinkage of the vertebral endplates), especially in the softer, cancellous bone regions (porous bone part).

2) Pore size: What is the best pore size for bone and vascular ingrowth?

- Ideal pore size ranges from 300 to 500µm.
- The size range supports vascularisation (process of blood vessels forming and growing in a tissue or organ), nutrient transport, and bone cell migration.
- It also promotes optimal osteoconduction (ability of a material to support the ingrowth of new bone tissue), encouraging new bone to grow into and around the cage structure.

3) What should the initial modulus be?

- The initial modulus should closely match the surrounding cancellous bone, typically 10-500 MPa.
- This avoids stress shielding, encourages load-sharing with native bone, and supports initial bone formation.
- Too high a modulus (like pure Ti) risks damaging adjacent bone; too low a modulus, risks insufficient mechanical support.

4) How can a change in the modulus be achieved?

- Through a composite cage design:
- Use a resorbable inner core (e.g., Mg alloy) with time-dependent degradation.
- Combine with a permanent outer layer (e.g., Ti or Si₃N₄) to maintain mechanical integrity.
- Use graded porosity or lattice structures that evolve the mechanical behaviour over time.
- Coatings (like chitosan) can control degradation rates, allowing the modulus to change in line with bone healing.

5) When should the change in modulus occur?

- After initial bone ingrowth begins, typically within 6-12 weeks post-operation.
- Modulus should gradually shift as new bone forms, allowing progressive load transfer from cage to bone, aligning with natural bone healing stages.
- The change should be synchronised with osteogenesis, not before, to avoid loss of structural support.

6) How can CT artefact be reduced if using a radiopaque material (e.g Ti)?

- Use low-density or structured Ti (e.g., open lattice, porous forms) to reduce beam hardening.
- Minimise metal volume in imaging-critical zones.
- Use coatings or alternative formulations (e.g., Ti-Al-V alloys) that lower radiodensity slightly.
- Design orientation can also be optimised to reduce artefact in the scan plane.

7) At what point is it most important that the artefact is low?

- During healing and monitoring phase, typically post-op weeks 6-12, when clinicians need to assess bone formation and fusion progress.
- High artefact during this period could obscure visibility of new bone, delaying intervention if healing is insufficient.

8) How can this be achieved through cage design?

- Design a radiolucent inner structure (e.g., Mg alloy or coated PEEK) that resorbs to improve imaging clarity over time.

- Use thin, porous metal sections rather than solid masses.
- Incorporate open chambers or windows that allow direct view of fusion zones.
- Add modular inserts that degrade and enhance radiolucency progressively.

9) How can material choice and design ensure good tissue integration?

Material choice:

- Ti with nanoscale surface features improves osseointegration.
- Mg alloys offer osteoconductivity and resorbability.
- PEEK coated with HA or nano-Ti balances mechanical properties and biological response.

Design Features:

- Use concave and open pores to promote bone growth.
- Introduce a permanent surface charge to attract proteins and growth factors.
- Use rhBMP-2 or nanoscale topographies to promote active bone regeneration.
- Consider piezoelectric material (e.g., PVDF) to stimulate bone under cyclic loads.

Manufacturing Process (a group's notes):

1) Peek uses FDM

- Good scalability: Ideal for customised cages as patients have diff bone geometries
- Simple design: Only need to manufacture the outer ring which has a simple design
- Affordable: For cage to be commercialised, price should be affordable, so trade-off was achieved by having a low manufacturing cost since material (PEEK) is expensive
- Rapid prototyping: Good for customisable cages since you can print and refine more cages quickly - might also help with pre-surgical planning

2) Az61 Mg alloy uses SLM

- Complex geometry: Lattice mesh has complex design and since it creates fully integrated geometries, you can make the inner core and mesh together in one build (reduces the need for joining process of inner core and mesh)
- High precision: Allows control over parameters such as laser width and speed - good for manufacturing of mesh which varies in thickness and to control the porosity of Mg alloy itself since diff patients will have diff bone mineral densities and therefore slightly diff elastic modulus of bone
- On demand manufacturing: ideal for customisable cages - implant is not bulk manufactured and is manufactured based on each patient

Material efficiency: Manufactured from powdered Mg alloy which are selectively melted and fused so excess sieved metal powder can be reused, cutting down costs

3) Dip coating for coating:

- Cost effective: Doesn't require vacuum systems or a lot of energy unlike other processes (plasma spraying etc.)
- Thickness control: You can control the viscosity of coating and the withdrawal rate as required)
- Uniform coating on complex shapes: Important that there is uniform coating throughout cage to control degradation rate of Mg alloy.

Case Study 4–Material Selection for a Ferritic/Austenitic Stainless Transition Weld with a Proposal for a Non-destructive Testing method to ensure Volumetric Weld Integrity

Objective: To select a suitable material and a non-destructive testing method to ensure a strong, flaw-free **weld** between ferritic/austenitic stainless steels.

Rules: The weld metal should be at least equal in strength and corrosion properties to the poorest component in the joint.

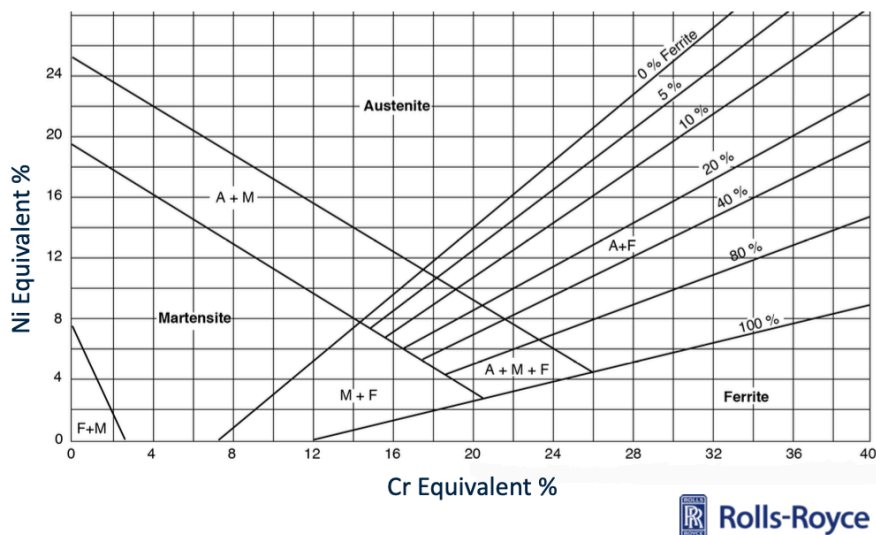
The microstructure must be resistant to cracking and Ferrite Number (FN) needs to be considered in relation to this; this is one by using Schaeffler diagram.

The chemistry between the weld filler and associated parent material(s) can be determined by using FN.

The chemical composition of the weld is likely to mean that it will be austenitic.

Corrosion is an important factor, as these metals are going to be operated at very high temperature and pressure.

Schaeffler Diagram



Material Selection Criteria:

Parent material(s) selection:

- Select a ferritic carbon alloy with:
 - Yield > 550 MPa
 - UTS = 650-950 MPa
 - Elongation > 15%
- Select an austenitic stainless steel with:

Yield > 200 MPa
UTS > 550 MPa
Elongation > 25%

- Select filler metal that is appropriate for the welding process you select with a nominal UTS of > 500 MPa.

NDE(non-destructive evaluation) and its importance to this case study

- NDE are known by NDT(non-destructive testing), NDI(non-destructive inspection), and non-destructive examination.
- These processes include testing, inspection, and examination, used to determine whether the object contains irregularities, discontinuities, or flaws (smt that is questionable or of interest in a component)
- A flaw that has been evaluated as rejectable is usually termed as a defect.
- NDE process do not change the physical added value to the component being tested, it is simply a way to test for safety in critical applications by ensuring that the tested component is fit for service and free from critical defects.

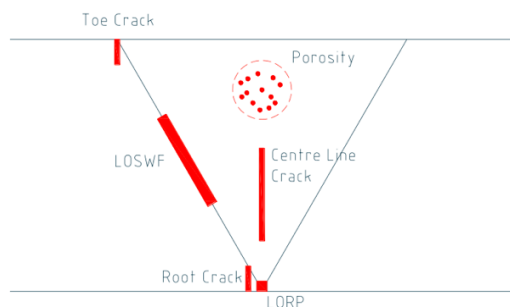
How do we select NDE methods?

- 1) Flaw detection and evaluation (PRIMARY FOCUS in case study)
- 2) Location determination and evaluation (PRIMARY FOCUS in case study)

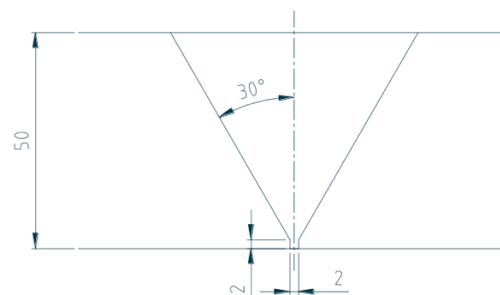
NDE Method Selection Criteria:

- Lack of root penetration (LORP)
- Lack of side wall fusion (LOSWF)
- Root Crack
- Toe Crack
- Centre Line Crack
- Porosity

Weld Defects



Case Study Weld Dimensions



Case Study Requirements:

- 1) Propose an appropriate NDE method to detect defects by considering:
 - The NDE method needs to be volumetric inspection method that can locate and size the defects.

- The NDE method should be sensitive enough to detect surface defects.
- Propose specific techniques to detect each of the defects.

2) Determine chemical composition of parent ferritic, austenitic stainless steel and weld filler metal

- Low carbon alloy steel– SA 508 or similar
- Austenitic stainless steel– Type 305 or similar
- Weld filler– Type 309 or similar

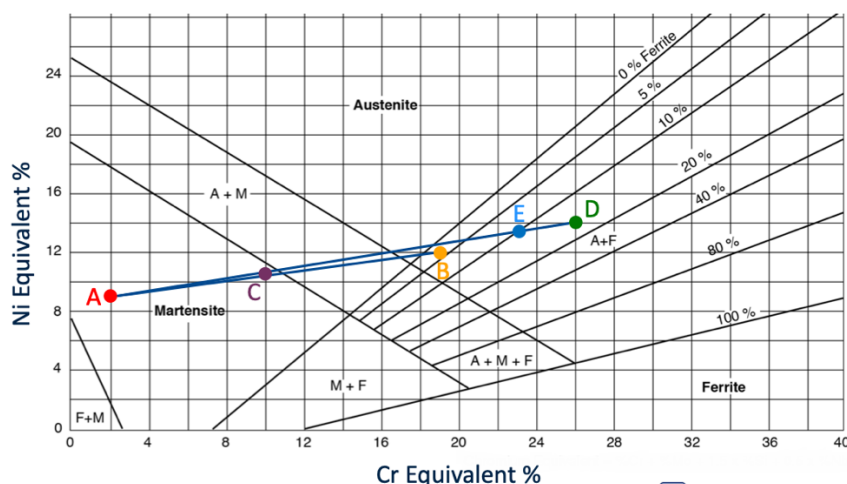
Material	C	Si	Mn	Ni	Cr	Mo	Nb	Cr Equiv	Ni Equiv
SA 508	0.19	0.1	0.25	3.4	1.6	0.45	0.01	2	9
Type 305	0.05	0.25	1.0	10	19.0	0.05	0.05	19	12
Type 309	0.01	0.65	1.0	12	25.0	0.25	0.01	26	13

Material	Yield	UTS	Elong
SA 508	650 MPa	820 MPa	22%
Type 305	260 MPa	585 MPa	40%
Type 309	400 MPa	590 MPa	40%

By using the calculate Cr Equiv and Ni Equiv, for each material type, these are important because these are the primary drivers for ferrite and ostene. Cr and Ni equivalent will determine the microstructure is most likely to be (will be ostomistic or to end up as a fluoric steel?)

- Point A = SA 508
- Point B = Type 305
- If no weld filler was used then the welded area will have composition of Point C.
- Point D = Type 309
- Point E = target composition of the weld which is controlled by dilution factor of the two parent metals and the weld filler.

Shaeffler Diagram



- Point C has a FN of 0, so is very weak crack resistance due to martensite formation.

- This is why an appropriate filler is used, Point D.
- Dilution of the weld filler with the parent material can be controlled to create Point E. E.g. in this case Point E would be ideal with FN=10-11 (acceptable range 5-20). Giving good balance of crack resistance and corrosion properties.

3) Select a volumetric NDE method to detect the weld defects described

There are 2 NDE method:

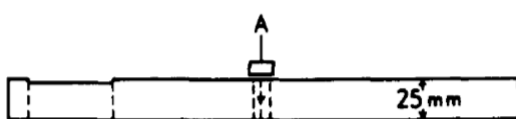
- 1) Radiographic testing (RT)
 - 2) Ultrasonic Testing (UT)
- UT is considered the more appropriate method due to it being superior to RT in detecting surface defects, such as a toe crack.
 - It is capable of detecting defects of variable orientations in terms of tilt angle and size.
 - It utilises the propagation of sound waves in the object or material tested. As such, the material properties such as elasticity is critical in determining velocity which must be known in order to locate and size defects appropriately.

4) Ultrasonic testing:

- 1) Longitudinal waves: Produced by a voltage applied to a piezo-electric transducer. The oscillation occurs in the direction of wave propagation.
- 2) Transverse waves: particles oscillate at a right angle to the direction of propagation. Shear waves require an acoustically solid material for effective propagation, and therefore, are not effectively propagated in materials such as liquids or gasses.

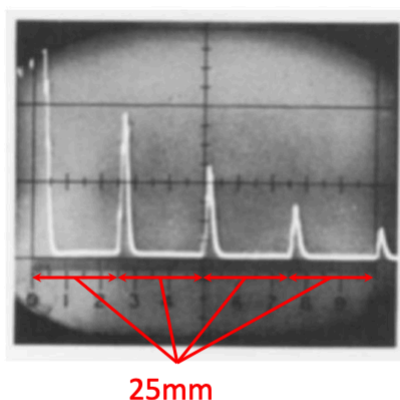
Ultrasonic Testing

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Position of probe A gives the 0-100mm thickness range.

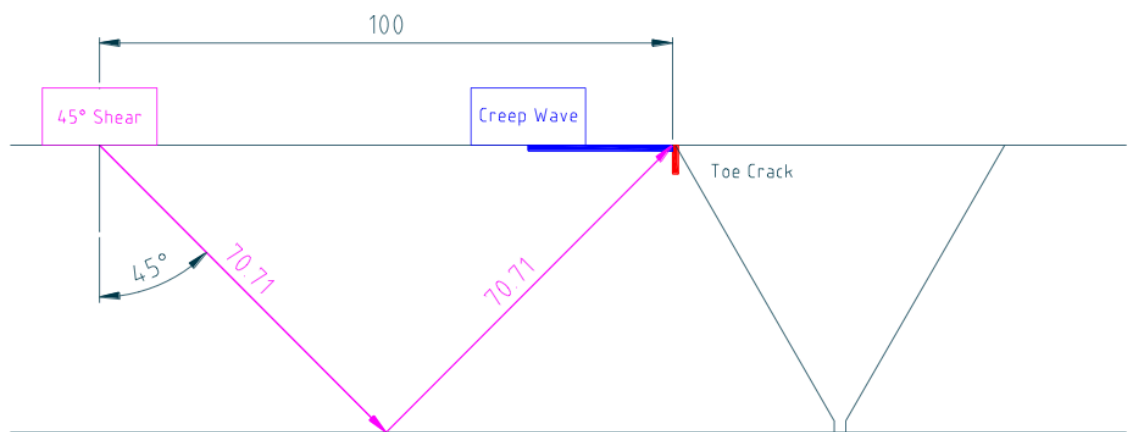
- The figure above shows a compression probe firing longitudinal waves through a 25mm thick component.
- The A-scan on the right shows the reflection of sound at the “backwall” or far side.



- The signals are produced in the A-scan figure because they represent an interface reflection from the far side of the component.
- The time base (y-axis) is calibrated with regards to the material velocity. So, each repeat signal occurs at 25 mm increments because the thickness of the component is 25 mm.
- Each subsequent signal decreases in amplitude because they represent an additional return trip of the sound wave between the near and far surfaces.
- This is the most basic setup of how we use ultrasonic waves, in this case to measure thickness.

5) Detail individual techniques to detect each defect

Toe Crack Detection

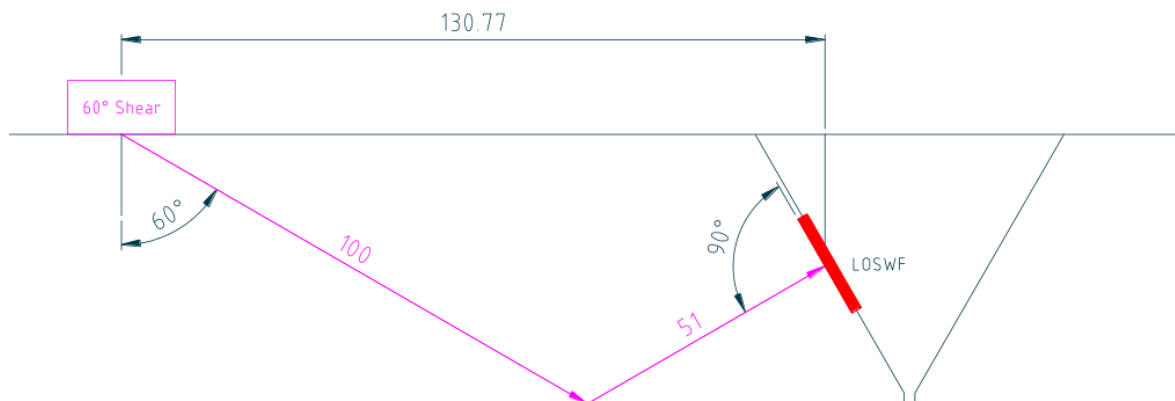


Two techniques could be used to detect a toe crack.

- 1) Creep Wave Probe - creep waves are used, which is a very high angle longitudinal wave (approx 78°) which is an effective surface wave
- 2) 45° Shear Probe - transverse waves are used from a 45° probe at "full skip".

LOSWF Detection

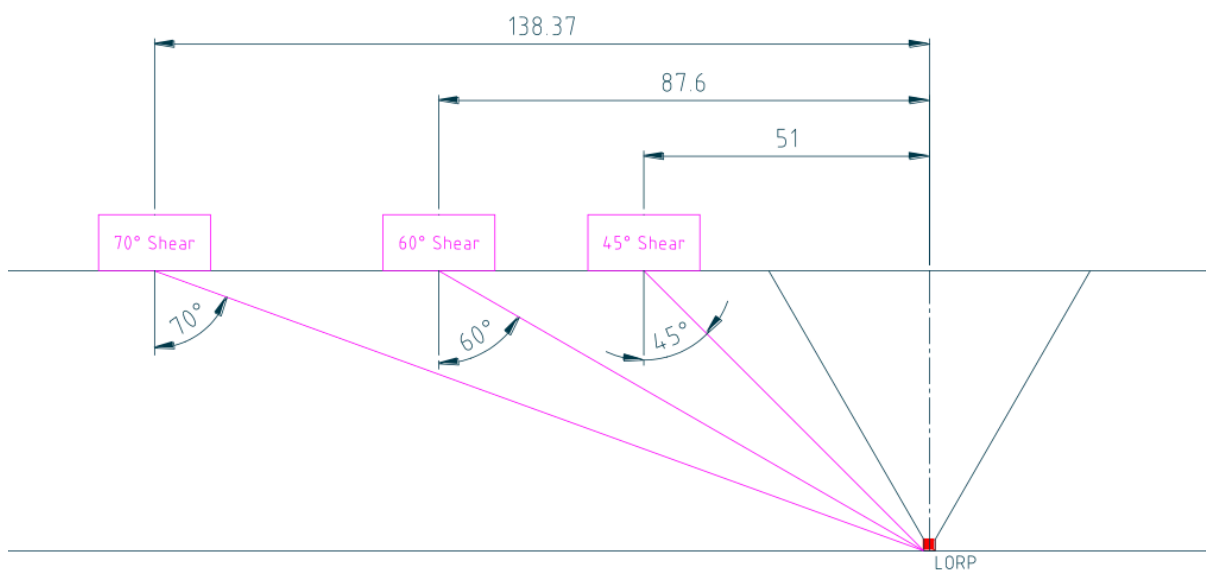
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Multiple angled shear wave probes will get a reflective response from a LOSWF defect. However, the angle which will give the best response is a 60° shear probe because the angle of the shear wave will strike the defect at 90° which will give a specular response.

LORP Detection

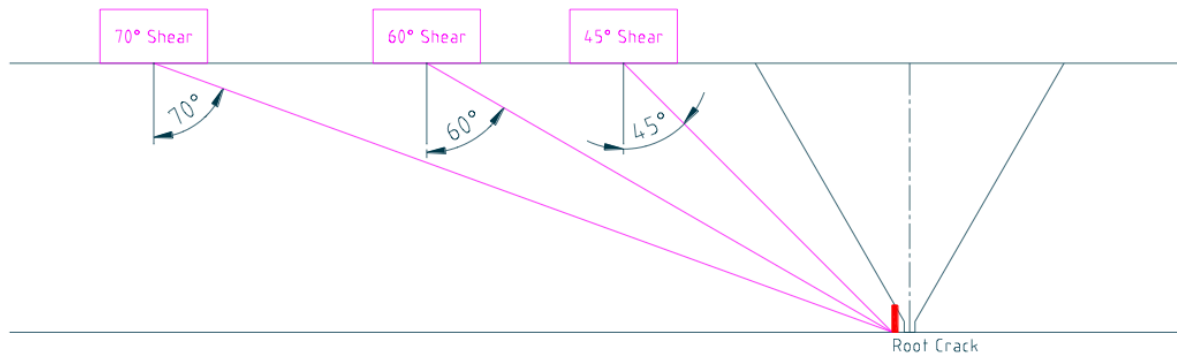
2



Multiple angled shear wave probes will get a reflective response from a LORP defect as shown above. Due to the nature of the location of such a defect, it should be noted that the defect can be found on either side of the weld centreline.

Root Crack Detection

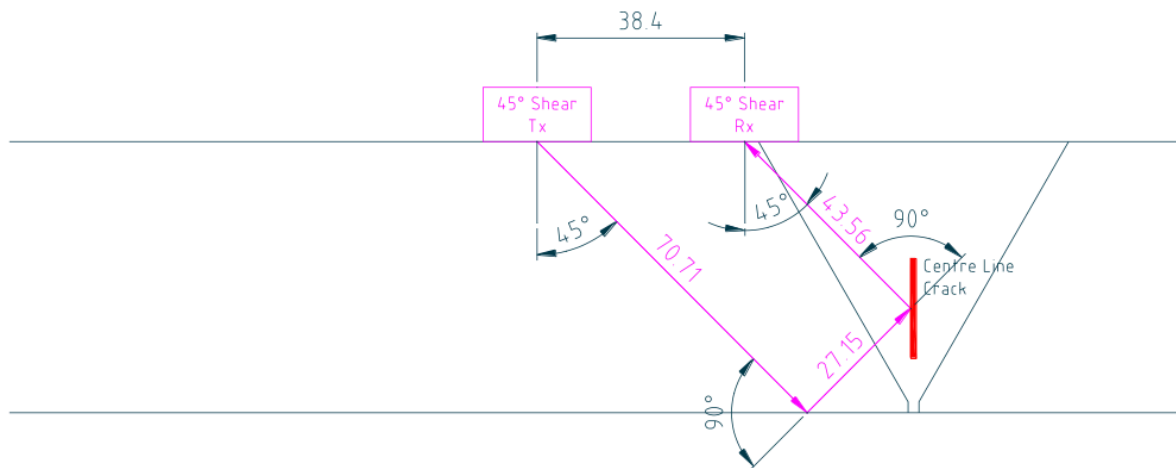
23



Multiple angled shear wave probes will get a reflective response from a root which is similar to a LORP as shown above. The main difference between a root crack a LORP will be its echodynamic response, which is how the amplitude changes when the probe traverses along the top scanning surface. A LORP will give a large response with a very well defined rise and fall on the timebase when viewed on an A-scan. A crack response will not and seem to have multiple peaks and troughs as would be expected from a relatively rough surface.

Centreline Crack Detection

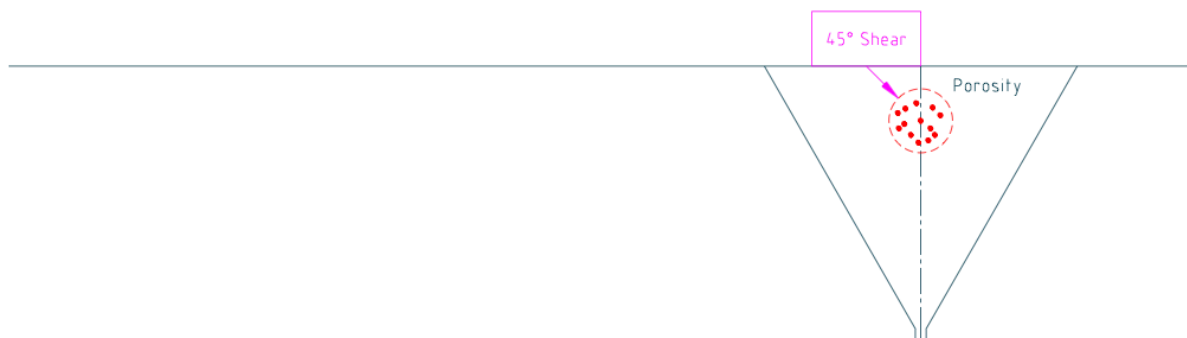
1



A centreline crack will not give a specular response from any angled probe. Therefore, a tandem setup with two probes is required. Tx is the transmitting probe and Rx is the receiving probe.

Porosity Detection

2



Multiple angled shear wave probes can get a reflective response from porosity cluster. Since the weld cap in this example is machined and accessible by the UT probe, a direct response is possible without skipping off the back surface.

Case Study 5–Biomedical Materials for Genicular Artery Embolization to Treat Pain from Knee Osteoarthritis

Problem: KOA (knee osteoarthritis) occurs when the cartilage in the knee wears down, causing pain to the patient. It is a long-term condition that can't be cured, but its symptoms can be treated.

Cause of pain: Causes inflammation in the joint due to mechanical damage. Resulting in pro-inflammatory factors that cause new blood vessel growth. Within the new blood vessels, new sensory nerves grow and invade the joint space (cartilage between knee bones (neoinnervation)). So moving the knee triggers pain signals via the new nerves.

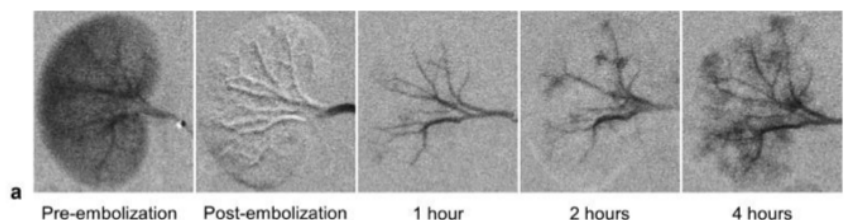
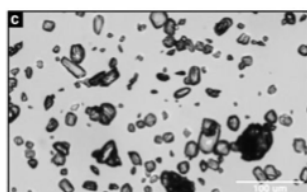
Treatment: One way is to kill these new nerves.

What is GAE (Genicular Artery embolization)?

- Embolisation is the intentional blocking of a blood vessel for therapeutic purposes.
- It is performed by an interventional radiologist under an image-guidance, using minimal invasive techniques.
- GAE is a new treatment where the arteries that feed the neovessels within the joint space are used to introduce particulate embolic agents.
- These block the flow, create ischaemia and kill the blood vessels and associated nerves, reducing the pain signalling.
- It is minimally-invasive

To carry out GAE, we need an embolic agent, so which?

- The best clinical data are from Japan, where the procedure is performed using a **mixture of antibiotics that form particles when mixed with contrast agents**.
- The particles vary in size depending on how they are produced and dissolve away over a few hours, creating a transient ischaemia (short period of symptoms–similar to stroke), but allowing blood flow to recover.
- This approach off-label and concerns over antibiotic use and allergic reactions means **this agent is not suitable for long-term adoption in medical practice**.
- An alternative device with similar properties is desired for mainstream GAE practice.



Solution:

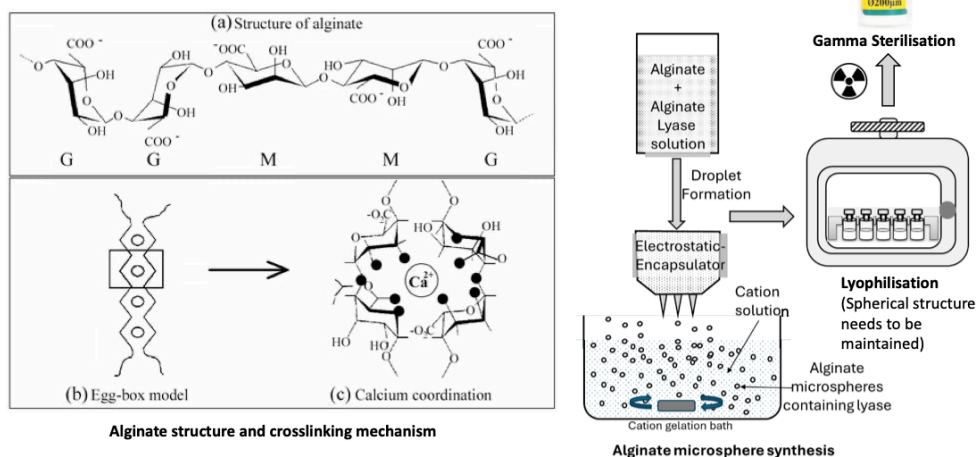
Material selection for embolic agent:

- **Alginate sourced from seaweed:** A commonly used, **highly biocompatible** material—safe for injection into the bloodstream when purified.
- **Formed into hydrogel microspheres** via calcium cross-linking (using electrostatic droplet generator, jet cutter, or microfluidics).
- **Size controlled to $200 \pm 50 \mu\text{m}$** , ideal for blocking small arteries during genicular artery embolization.
- **Embedded with alginate lyase enzyme**, which **controls the degradation rate** so the microspheres dissolve safely over time.
- **Inspired by Japan's Sakura (cherry blossom)** to reflect the **temporary**

Nature of the treatment, giving rise to the name **SakuraBead**:

- The **SakuraBead embolic agent is injected into the genicular arteries to temporarily block blood flow**, which causes **ischemia (oxygen deprivation)** in newly grown blood vessels and nerves inside the knee joint.
- This process **kills the abnormal nerves responsible for chronic pain in knee osteoarthritis, reducing pain** without the need for invasive surgery.
- After doing its job, the microspheres **safely degrade** into small fragments that the body can excrete—**leaving no permanent implant behind**.

Microsphere Manufacture



That is the alginate structure, you use calcium to cross-link it, you have controlling mechanisms because glucuronic acid residues are the things that bind the calcium. And you can control the M:G ratio that gives you control of cross link density.

Also, the molecular weight has an impact.

Manufacturing: You mix the alginate with the layers; you put it in an electrostatic encapsulator which drips into a calcium bath to form microspheres. You do that at low pH, because that inactivates the enzymes so that it doesn't eat itself during production.

Then you put it into a lyophiliser which is a freeze dryer, takes the water out, and that frozen the matrix so that the enzymes can't work on it. Then you get a dry powder, that can't degrade itself because there's no water in there. SO that's how you stabilise it and then you can gamma radiate it, so that it stabilises without breaking down.

Compressibility and Isobouyancy Explained

Compressibility:

Elastic recovery is important, must have ability to deform dramatically and recover.

- **Hydrogel microspheres (MS)** used in embolization are **soft and flexible**, which allows them to **squeeze through very narrow tubes** (called microcatheters) during injection.
- Even when compressed, they are **strong enough not to break** because they are **viscoelastic** – they can **stretch or deform**, then **bounce back to their original shape** after the pressure is removed.
- This property is important because **once inside the blood vessel**, the microspheres need to **fully return to shape** to **properly block the blood flow**.
- If they **don't recover their shape**, they may **fail to block the vessel** properly, which has happened in other products and leads to **poor treatment results**.

Isobouyancy (Suspension Ability):

- Hydrogel microspheres are also **light enough to stay suspended** in the **contrast liquid** used during injection.
- This means they don't **sink too quickly**, allowing for a **smooth and controlled injection**.
- If the microspheres **settle out too fast**, it becomes hard to deliver them evenly, which affects how well the treatment works.

Degradation of the Embolic Agent (SakuraBead)

How it Works:

- **Alginate** (the main material in SakuraBeads) **doesn't naturally break down in the body** because the body lacks the enzyme to digest it.
- So, they **embed a special enzyme** called **alginate lyase inside the bead**. This enzyme is **only active on the bead itself**.

Controlling the Breakdown Time:

- The **speed of degradation** can be **adjusted** by changing:
 - The **molecular weight (MW)** of the alginate,

- The **ratio** of its building blocks (M:G),
 - The **amount of enzyme** inside the bead.
- This lets developers **set how long the bead lasts** – anywhere from a **few minutes to several weeks**, depending on what's needed.

What Happens During Degradation:

- The enzyme **cuts the alginate into smaller pieces**, lowering its **molecular weight**.
- When the pieces are small enough, they can **pass through the kidneys** and be **safely excreted** from the body – this is called **renal clearance**.

✓ Key Advantage:

- The **breakdown doesn't depend on the body's response**, like it does with materials such as **gelatine or starch**.
- That means the **timing is predictable and reliable**, no matter where the bead is placed or how a patient's body reacts.

Colour and Radiopacity:

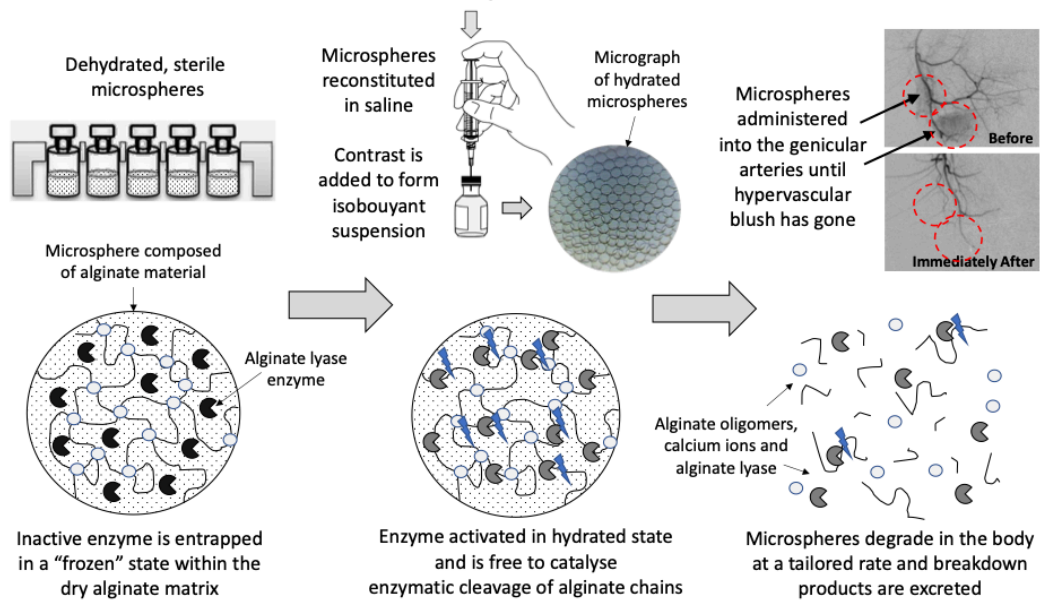
Colour and Radiopacity

- Permanent embolic particles are often dyed with a colouring agent so that they can be seen to form an even suspension for delivery
- This is more complex for a degradable embolic as the developer must demonstrate what happens to the dye in the body and that it is safe
- Radiopaque microspheres have been made by adding barium or tantalum particles, or by adding iodinated species to the MS polymer structure
- A lot is required to be visible (coating is insufficient) and the MS become very dense and difficult to deliver; in the GA they would be a challenge to visualise



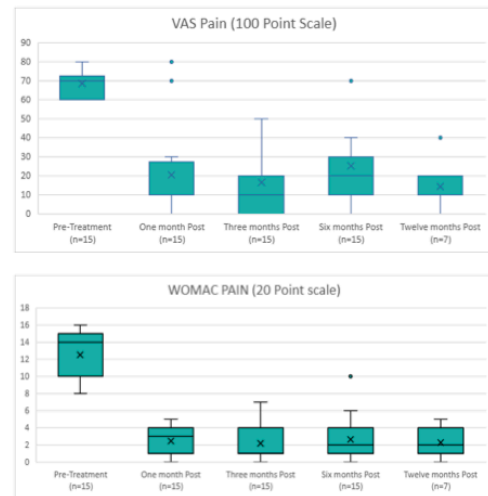
Preparation for use of SakuraBead

SakuraBead®: Preparation for Use



SakuraBead: First in Human Pain Relief Study

- 15 Patients treated with KL grade 2/3 KOA
- The average volume of SakuraBead suspension injected to reach the desired embolisation endpoint was 1.1 ml (a range of 0.4-1.9 mL)
- An average number of 2.4 genicular arteries were embolised per patient
- GAE was achieved in all the patients with 100% technical success
- Significant pain reductions seen post-procedure (14/15 patients) and sustained for at least 12 months



13

SakuraBead: A Combination Product

- Alginate microspheres are introduced into the genicular arteries and temporarily block the blood flow, inducing ischaemia which kills neovessels and neonerves in the joint space:
 - Primary mode of action: PHYSICAL EFFECT – DEVICE
- Alginate lyase enzyme inside the beads is activated once reconstituted and acts solely on the beads to degrade them into small fragments that can be excreted, independent of any degradative mechanism in the body. The presence and concentration of the enzyme control the rate and degree of degradation
 - Secondary action: BIOLOGICAL EFFECT – BIOLOGIC

FDA Classification: Device-Biologic Combination Product

14

Your Challenge:

- 1) Using your knowledge of materials science, select a material/ combination of materials that could meet the considerations for use as a temporary embolization agent for GAE (geniculate artery embolization).

- 2) There are several materials being evaluated in the clinic for GAE, they are not necessarily the best options and each will have its pros and cons. Try to come up with a selection that addresses the weakness of what's already out there.
- 3) Justify your material choice by demonstrating how the material's properties address each of the key performance attributes of a temporary embolic agent.

Some Considerations:

- 4) This is a medical device that is injected into the blood stream and needs to be safe.
- 5) It needs to be delivered using the conventional methods appropriate for embolization.
- 6) It will require mechanical characteristics to work as an embolic agent.
- 7) It needs to disappear over a specified time period-how will you achieve this and what happens to the material as it breaks down?
- 8) How will it be manufactured?

Link:

https://qmplus.qmul.ac.uk/pluginfile.php/4644605/mod_resource/content/3/%235-Andy-Solution.pdf

Model answers:

1) What material or combination of materials would you choose as a temporary embolic agent for GAE, and why?

Material chosen:

- Alginate hydrogel microspheres with embedded alginate lyase enzyme.
- Alginate is biocompatible, especially when highly purified.
- Cross-linked with calcium, alginate forms hydrogel beads, ideal for microcatheter delivery.
- Alginate lyase inside the bead enables controlled, predictable degradation, unlike other materials that rely on slow hydrolysis or the patient's biology.

2) What are the key mechanical and biological performance requirements for the material, and how does your choice meet them?

- **Particle size:** Controlled at 200 (+-) 50µm, ideal for GAE.
- **Mechanical strength:** Viscoelastic hydrogel structure is compressible and recover shape, preventing rupture or migration.
- **Isobouyancy:** Stable suspension in contrast media; no premature settling.
- **Biocompatibility:** Alginate and its breakdown products are non-toxic and excretable.
- **Controlled degradation:** Tuned from minutes to weeks via enzyme concentration and alginate composition.

3) How will the material be delivered using conventional embolization techniques?

- Microspheres are injected via microcatheters, designed to not block small-bore systems due to their deformability.

- Their compressibility and isobouyancy help ensure smooth, even delivery with minimal sedimentation.
- Beads recover their shape after delivery, ensuring they remain in target vessels.

4) How do you ensure that the material degrades safely over time, and what happens to it after degradation?

- degradation is controlled by alginate lyase embedded in the beads.
- This enzyme cleaves alginate chains into fragments small enough for renal clearance.
- Rate of degradation is tuneable, not reliant on patient-specific conditions (unlike gelatine or starch).
- Safe breakdown: no systemic enzyme activity, no inflammation.

5) How will your material minimise inflammation or immune response after injection?

- Alginate is highly biocompatible, especially when purified to remove residual proteins.
- The enzymes only act within the bead, avoiding off-target effects.
- No immune-reactive components are introduced into the body from the material or its byproducts

6) How will the embolic agent be manufactured at scale, and what form will it take?

- Manufactured as sterile, off-the shelf microspheres with a 3-year shelf-life.
- Microspheres formed using methods like electrostatic droplet generators, microfluidics, or jet cutting.
- Beads contain pre-measured alginate lyase to ensure reproducibility in degradation rate.

7) How does your solution align with regulatory requirements and clinical safety expectations for temporary embolic agents?

- Temporary embolization is safer and preferred by regulators compared to permanent implants.
- The first-in-human study with SakuraBead showed:
 - 1) Around 90% technical success
 - 2) Pain reduction in 14/15 patients, lasting over 12 months
- Alginate lyase is device-specific and avoids systemic exposure, supporting a controlled and traceable degradation mechanism.

Case Study 6–Cam Damper Masses for F1 Power Unit

Objective:

- Find a solution that will produce the least wear on the damper masses or carousel while matching the damping torque and vibration order.

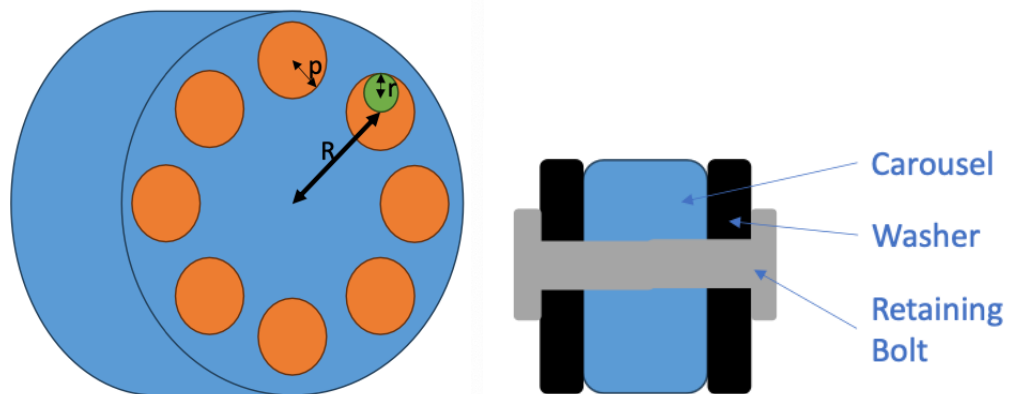
What are rotational dampers?

Dampers reduce object oscillations by opposing their vibrational movements.

- Rotational dampers oppose vibrations in a torsional system to reduce the fatigue on components.
- The less the component moved back and forth the longer it will last, and the smoother the movement in the power unit will be.

What are pendulum dampers?

- Oppose the movement of rotational dampers, with a high-density weight that swing in the opposite direction to motion.
- E.g.: the carousel (blue) rotates clockwise, the damper mass (green) will rotate anticlockwise in the bore (orange).



The mass is held in the bores by two washers on either end. The **washers** and **carousel** are made of the **same material**.

So, when **considering wear**, you should consider **your chosen material interacting** with **EN9 steel**.

Consider both yield strength and hardness.

Constraints:

- Material must be **below a density of 18,100 kg/m³** (desirable to get as close to 18,100).
- Packaging of the carousel is **restricted to 18 mm diametrically**
- May fit as many pendulum dampers as you like per carousel
- The **camshaft carousel** and **washer** to assemble the system are made of **EN9 steel**.
- **Least wear on the damper mass or carousel**

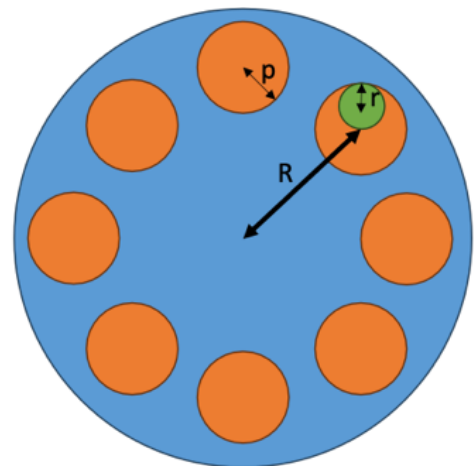
- Operating in a power unit– exposed to temperatures up to 140-160 Celsius, So its **minimum maximum temperature should be at least around 140-160** degrees. It will be **exposed to oil and vibration within the carousel.**
- **The point of the carousel is that they create the damping effect only when rolling on the outside. So we will be concerned about not damping the system when:**

The damper is too dense (too heavy), as the only thing supporting it on the outside of the carousel is centrifugal force, therefore when starting the engine or at low RMP, they're going to be vibrating around. So, you would want them to have at least some tensile strength (something that won't have them fatigue). As the more they vibrate, the more they start to 'fatigue' – damage.

Constraints	Objectives	Free Variables
Density < 18,100 kg/m ³ Material types i.e. must be a solid Hardness around EN9 steel (100 < HV < 400) Max Temperature > 140 degC Corrosion resistant to oil	Wear Resistance – high ((Hardness * Contact Area) / (Sliding Velocity * Load)) Hardness – high Density – high Yield Strength – high	Cost

Calculations:

- Damper Mass is a simple cylinder
 - Radius = designable
 - Length = 15mm
- Damper Torque:
 - $$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$
 - **Your damper torque should equal as close to 19.67Nm @ 11000rpm**
- Vibration Order:
 - $$\left(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))} \right)^{1/2}$$
 - **The vibration order of this system is 4**



m = mass of damper
 ω_{engine} = engine speed
 R = carousel radius
 $e = p - r$
 r = roller radius
 p = bore radius
 $n_{dampers}$ = number of dampers (and bores)

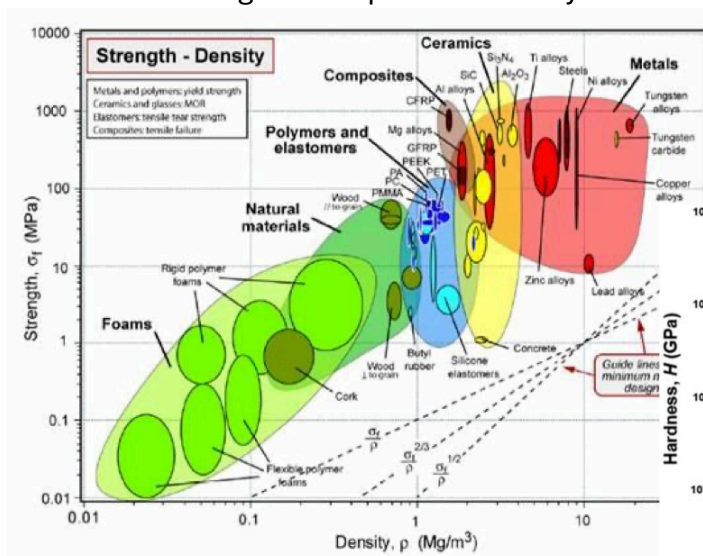
$$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$

$$(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))})^{1/2}$$

Variable	Damper Torque	Vibration Order
Mass – higher	Higher	-
Bore Radius – higher	Higher	Higher
Roller Radius – higher	Lower	Lower (order of 2)
Number of dampers – Higher	Higher	-

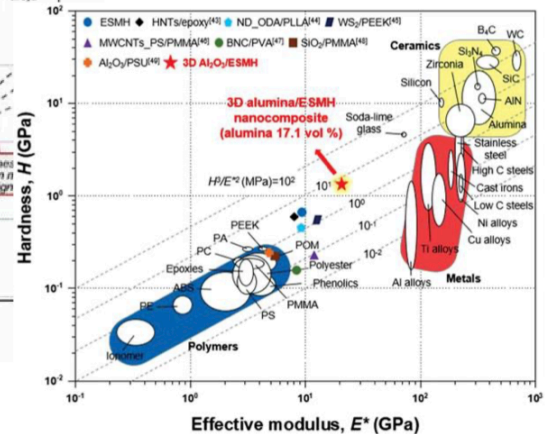
Material Selection:

- Use materials with higher wear resistance for the rolling/sliding components.
- Applying coatings such as chrome plating or specialised wear-resistant coatings can help reduce friction protect surfaced from wear.
- E.g.: heat treatments, anodising or PVD/CVD (DLC-Diamond-Like Carbon) coatings can improve durability.



Options:

Tungsten Alloys
Tungsten Carbide
Nickel Alloys
Steel



Materials chosen:

Material	Density (kg/m ³)	Hardness (HV)	Yield Strength (MPa)	Cost (£ per kg)
Tungsten Alloys	17000-18500	300-400	500-1000	75-230
Tungsten Carbide	14500	1600-2500	300-350	200-300
Nickel Alloys	8000-9000	100-400	150-170	40-75
Steel	7000-8000	100-800	200-500	100-200

- **Tungsten alloy**, densest material you can get, most effective damper, and able to roll the most.

Lubrication:

- Ensure that proper lubrication is applied to the moving components. This helps reduce wear by minimising direct contact between moving parts.

Friction control:

- Ensure that the damper is designed so that there is enough clearance between components.
- Surface finish of materials will make a huge difference; therefore machining should be considered for both carousel and damper mass.

Link: https://qmplus.qmul.ac.uk/pluginfile.php/4627343/mod_resource/content/1/6-Cara-Feedback.pdf

Model answers

1) What is a damper mass?

Dampers reduce object oscillations by opposing their vibrational movements.

2) How does it work?

A damper mass works by moving in the opposite direction of vibrations, using forces like friction or fluid resistance to absorb energy and reduce oscillations—sometimes affected by centrifugal or centripetal forces in rotating systems.

3) How should you choose the materials for the damper mass and the carousel that it sits in?

Should be a material with high density, great durability and fatigue resistance and corrosion resistant.

- Strong but lightweight (Ashby chart: Density vs Strength)
- Durable and fatigue resistant (Young's modulus vs Density)
- Corrosion resistance

4) What materials should you use and explain why?

Material	Density (kg/m ³)	Hardness (HV)	Yield Strength (MPa)	Cost (£ per kg)
Tungsten Alloys	17000-18500	300-400	500-1000	75-230
Tungsten Carbide	14500	1600-2500	300-350	200-300
Nickel Alloys	8000-9000	100-400	150-170	40-75
Steel	7000-8000	100-800	200-500	100-200

Tungsten Alloy scoring the best for strong and lightweight, durable and fatigue resistant as well as better resistance to corrosion.

5) How might you make an individual damper mass?

Can be made using CNC (Computer numerical control) machining of a dense metal like tungsten, chosen for its high density and stiffness, to precisely achieve the required inertial and damping characteristics.

6) How should we surface treat the damper mass to improve lifetime?

Lubrication:

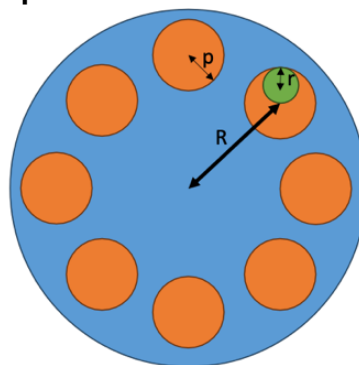
- Ensure that proper lubrication is applied to the moving components. This helps reduce wear by minimising direct contact between moving parts.

Friction control:

- Ensure that the damper is designed so that there is enough clearance between components.
- Surface finish of materials will make a huge difference; therefore, machining should be considered for both carousel and damper mass.

7) How do the vibration and torque calculations work?

- Damper Mass is a simple cylinder
 - Radius = designable
 - Length = 15mm
- Damper Torque:
 - $$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$
 - Your damper torque should equal as close to **19.67Nm @ 11000rpm**
- Vibration Order:
 - $$\left(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))} \right)^{1/2}$$
 - The vibration order of this system is 4



m = mass of damper
 ω_{engine} = engine speed
 R = carousel radius
 $e = p - r$
 r = roller radius
 p = bore radius
 $n_{dampers}$ = number of dampers (and bores)

$$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$

$$(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))})^{1/2}$$

Variable	Damper Torque	Vibration Order
Mass – higher	Higher	-
Bore Radius – higher	Higher	Higher
Roller Radius – higher	Lower	Lower (order of 2)
Number of dampers – Higher	Higher	-

8) How do you design to counter fretting/ abrasion/ wear in practice?

- Use wear-resistance materials like tungsten alloy for durability.
- Apply proper lubrication to reduce friction and wear
- Design with enough clearance to minimise fretting (when two contacting surfaces experience small, repetitive motion–the damper mass moving around the carousel) from micro-movements.
- Ensure smooth surface finish through precision machining to reduce abrasion